

Sai Sahasra Peddabuddi

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EDUCATION

Rutgers University

New Brunswick, NJ

Bachelor of Science in Computer Science; Minor in Business Administration

Expected May 2028

- **Relevant Coursework:** Data Structures, Computer Architecture, Discrete Structures I & II, Data Management for Data Science, Introduction to Artificial Intelligence, Operations Management, Financial Accounting
- **Leadership & Programs:** RBS Student Ambassador (Dean's Office); Peer Mentor; Capital One Tech Summit (2nd of 9); SIG Discovery Day; Accenture Extern

PROFESSIONAL EXPERIENCE

Travelers

Hartford, CT

Software Engineer Intern

Jun 2026 – Aug 2026

- Engineered a GenAI assistant powered by an internal Travelers AI model, giving a **40-person** underwriting team on demand summaries of account and policy details from historical BI Workstation data
- Shipped **8** production features in **JavaScript** for the Risk Evaluation platform, integrating **3** backend microservices to power real time commercial risk assessment for underwriters
- Presented feature work at a sprint demo to **250+** engineers and business stakeholders across the Business Insurance organization

Travelers

Hartford, CT

Software Engineer Intern

Jun 2025 – Aug 2025

- Engineered an end-to-end **Kafka** to **ServiceNow** pipeline in **TypeScript** and **Java**, automating creation of **100+** tasks for the Business Insurance Work Orchestration team during the pilot
- Architected **10** REST APIs powering the pipeline, validating endpoints in **Postman** and writing **SQL** to reconcile generated **ServiceNow** tasks against source **Kafka** events
- Presented the integration to **250+** people across the Business Insurance organization, framing the technical design for business stakeholders

Red Hat

Raleigh, NC

High School Internship Program

Jul 2024 – Aug 2024

- Selected as **1 of 6** from **2,000+** applicants for the inaugural cohort of Red Hat's High School Internship Program
- Built a website for future intern cohorts and pitched **35+** Red Hat directors and program leadership on continuing the pilot

PROJECTS

Limit Order Book Matching Engine | C++17, Make, Linux | github.com/SaiPedd/orderbook

2026

- Implemented a price-time priority matching engine in **C++17** sustaining **9.4M orders/sec** with **83ns** median and **292ns** p99 order-processing latency over a **1M-order** synthetic workload
- Implemented limit order crossing with partial fills, resting remainders, FIFO time priority, and cancellation, verified by a **20 case** suite covering multi-level sweeps and priority invariants
- Stored price levels in a **std::map** with per-level **std::deque** queues for **O(1)** FIFO fills, and indexed orders in an **unordered_map** for **O(1)** cancellation off the matching critical path

Ledger | React, TypeScript, Node.js/Express, SQLite | github.com/SaiPedd/Ledger-

2026

- Architected a full stack personal finance app in **React**, **TypeScript**, **Express**, and **SQLite**, powering **14** REST API endpoints across **4** tables for transactions, scenarios, lifestyle goals, and debts
- Built a what if scenario simulator projecting monthly cash and invested alternative balances under fixed market return and volatility assumptions, modeling expected, best, and worst case outcomes
- Built a debt payoff calculator simulating avalanche and snowball strategies month by month, computing months to debt free and total interest paid for each

TECHNICAL SKILLS

Languages: C++, Java, Python, JavaScript, TypeScript, SQL, HTML/CSS, Bash

Frameworks & Libraries: React, Node.js, Express, REST APIs

Tools: Git, Make, Linux, SQLite, Kafka, ServiceNow, Jira, Postman, Power BI, Excel, Tableau

Concepts: Data Structures & Algorithms, Systems Programming, Microservices, Concurrency, Benchmarking & Profiling, Database Design, Testing, Agile/SDLC